

Arvind Yogesh Suresh Babu

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Work Experience

University of Michigan, Ann Arbor, United States

September 2025 - Present

Student ML Researcher

- Built an automated perception to planning evaluation pipeline (ROS, Open3D, Python), reducing manual test effort by 70% and enabling batched A/B model comparisons.
- Trained a multi-task perception and short-horizon motion-forecast model (PyTorch) that cut 0-3s trajectory error by 15% and reduced collision-warning events by 9% in closed-loop simulations.
- Added distribution-shift monitoring and regression tests (MLflow, TensorBoard) to detect model degradation and speed rollbacks.

Vestas Wind Technology, Chennai, India

June 2024 - July 2025

Data Scientist Intern

- Developed gearbox-failure forecasting (SCADA + logs) using engineered time-series features and an LSTM and XGBoost ensemble; AUC 0.82 and median lead time about 18 days for alerts.
- Productionized streaming inference (Kafka / Spark / ONNX serving) with 80 ms median latency and automated retraining via Airflow, cutting unplanned downtime by 25%.
- Integrated spare-parts demand forecasting with the maintenance pipeline, lowering overstock by 15%.

Computer Society, MIT, Chennai, India

Mar 2021 - June 2024

ML Engineer

- Designed and productionized a unified semantic-search and RAG pipeline (dual-encoder retrieval + FAISS ANN and cross-encoder re-ranker) using sentence-transformers and LLM re-ranking; improved Top-1 relevance by 28% vs TF-IDF and achieved median query latency <120 ms. Applied knowledge distillation, INT8 quantization, and ONNX runtime optimizations to increase throughput 3x and cut CPU inference cost 2.5x.
- Built a production-style ADAS real-time perception pipeline (Kafka + Spark Structured Streaming) with a CUDA-ready FastAPI inference service (ssdlite MobileNetV3); optimized micro-batching and model path to reach 20 FPS and p95 end-to-end latency <150 ms in benchmarked runs.
- Developed Robotic Machining Analytics: 1D-CNN + GBDT ensemble for hole-quality prediction with Jetson-edge anomaly detection; deployed real-time alerts and MLflow-tracked experiments to accelerate defect triage.

Education

University of Michigan, Ann Arbor

Aug 2024 - Dec 2026

MS Data Science

CGPA: 3.87

Relevant Subjects: Applied Machine Learning, Natural Language Processing, Regression Analysis, Advanced DBMS

Madras Institute of Technology (MIT)

Aug 2020 - June 2024

BE Mechanical Engineering

CGPA: 3.90

Relevant Subjects: C Programming, Python Programming, Data Structures & Algorithms, Deep Learning, Software Engineering, Robotics Simulations

Publication

Attribution-Guided Masking for Robust Cross-Domain Sentiment Classification (link). arXiv:2605.03091 2026.

AGM is a gradient-based training loss that detects and penalizes highly attributed domain-specific tokens during fine-tuning.

Evaluated zero-shot across 4 domains (8 seeds); on Sentiment140 reported Delta: 0.244 vs DANN 0.264 DRO 0.248 Fish 0.247 IRM 0.238, and showed token-level attribution suppression for mentions/hashtags/slang.

Project Work

CLEVER, Cluster Level Eviction for Vector Embedding Retrieval | PyTorch, FAISS, HNSW, LSH, LMSYS (link)

Jan 2026 - Apr 2026

- Built a semantic caching and ANN benchmarking framework on LMSYS Chat 1M, 579753 unique queries, 384 dimensional embeddings, evaluating Flat, IVF, HNSW, and LSH under realistic LLM workloads.
- Benchmark results: HNSW recall 0.996 at 0.56 ms and 11056 QPS versus Flat recall 1.000 at 17.39 ms and 58 QPS, and routing achieved 59.1% hit rate with 80.3% semantic quality and 60.3% latency savings at threshold 0.76.

Pasupathy-ai | React, TypeScript, Flask, FAISS, AWS (link)

Oct 2025 - Dec 2025

- Built a RAG powered personal AI assistant with React 18 TypeScript, Flask, FAISS, and Google Gemini; deployed with Docker on AWS Elastic Beanstalk, with frontend on S3 plus CloudFront and production CI CD via GitHub Actions.
- Achieved documented production performance metrics: initial embedding build about 60 minutes, FAISS index load about 5 seconds, average query latency 2 to 5 seconds, backend memory about 2 GB, and index storage about 500 MB.

Spotify Music Recommender | LightFM, OpenL3, FAISS, Transformer Reranker, AWS (link)

May 2023 - Aug 2023

- Implemented a hybrid recommender combining LightFM collaborative filtering with OpenL3 audio embeddings and FAISS ANN for low-latency nearest-neighbor retrieval; validated with offline NDCG/Recall experiments.
- Added a session-aware Transformer reranker and an online personalization pipeline (AWS Lambda + DynamoDB); instrumented a Spotipy-based evaluation harness for playback A/B tests.

Skills

Languages: Python, SQL, JavaScript, TypeScript, C++ | **ML and Data:** PyTorch, scikit-learn, NumPy, Pandas, LLMs, feature engineering, PySpark | **Data and Retrieval:** ETL and ELT, data pipelines, FAISS, RAG pipelines, Hugging Face Transformers | **Cloud and Infrastructure:** AWS, Docker, Terraform, Azure, FastAPI, ONNX, Kafka, Spark | **Tools:** Git, MLflow, pytest, Linux, Tableau